

FERN v2

Foot Gesture Recognition Pipeline

Paper Readiness Report

Generated: Aug 29, 2026

8 Gesture Classes | 44 Subjects | CNN1D-BiLSTM-Attention

DATASET OVERVIEW

- 88 recording sessions from 44 unique participants
- Front-facing camera, 8 gesture classes
- MediaPipe Holistic: 10-joint skeleton, 30-dim vector (x,y,z)
- 30 fps extraction

GESTURE CLASSES

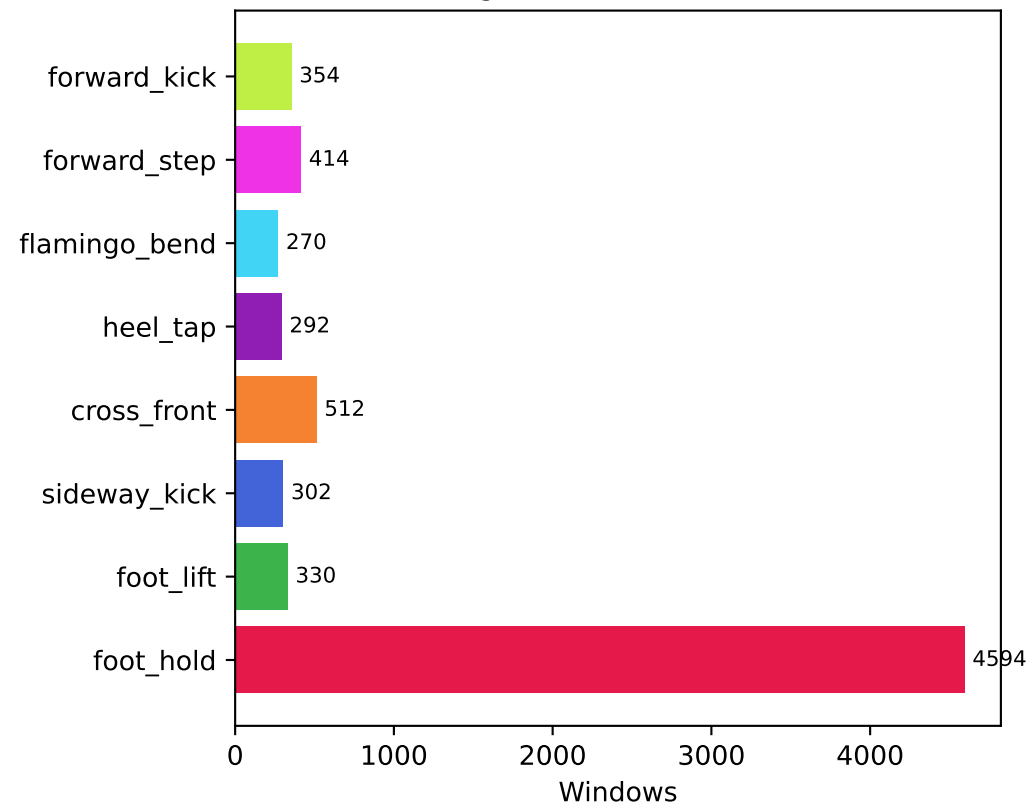
1. foot_hold – Static foot placement
2. foot_lift – Lift foot off ground
3. sideways_kick – Lateral kick motion
4. cross_front – Cross foot in front
5. heel_tap – Tap heel on ground
6. flamingo_bend – Single-leg bend (flamingo pose)
7. forward_step – Step forward
8. forward_kick – Kick forward

DATA SPLITS

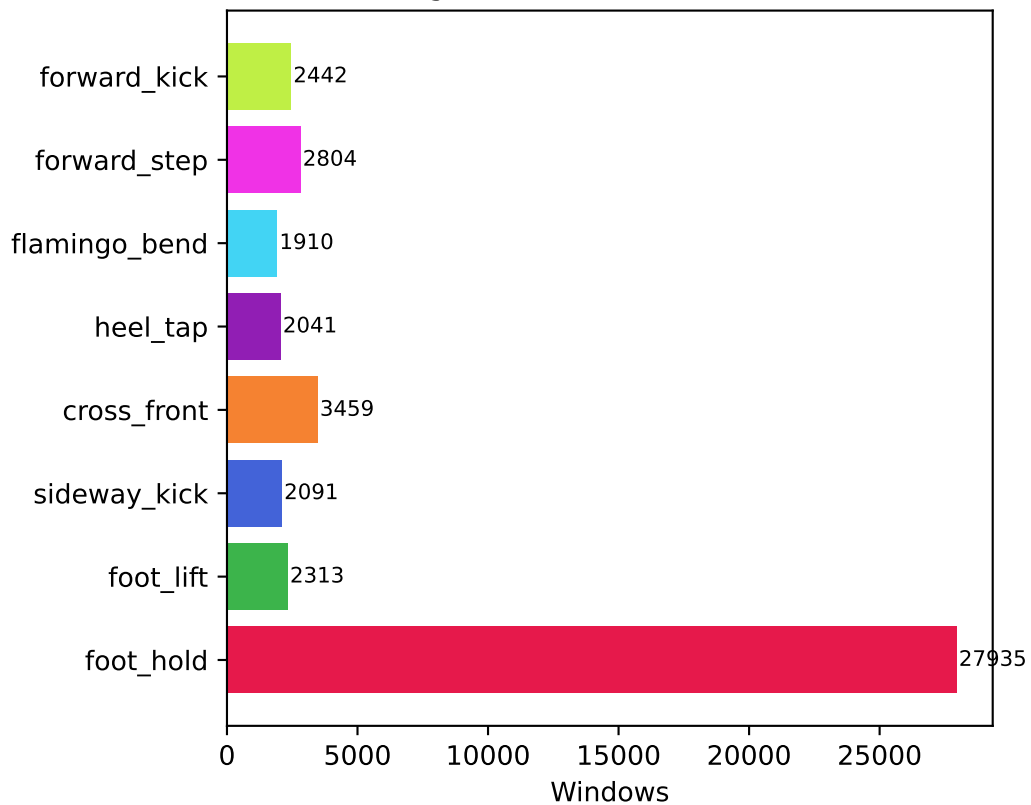
Original: 88 files (44 subjects × 2 mirror)
Augmented: 484 files (rotation, spatial, interpolation)
Held-out test: 12 files (6 subjects × 2 mirror)
Training subjects: 38 | Held-out: 6 (p25,p30,p35,p40,p45,p50)

Class Distribution

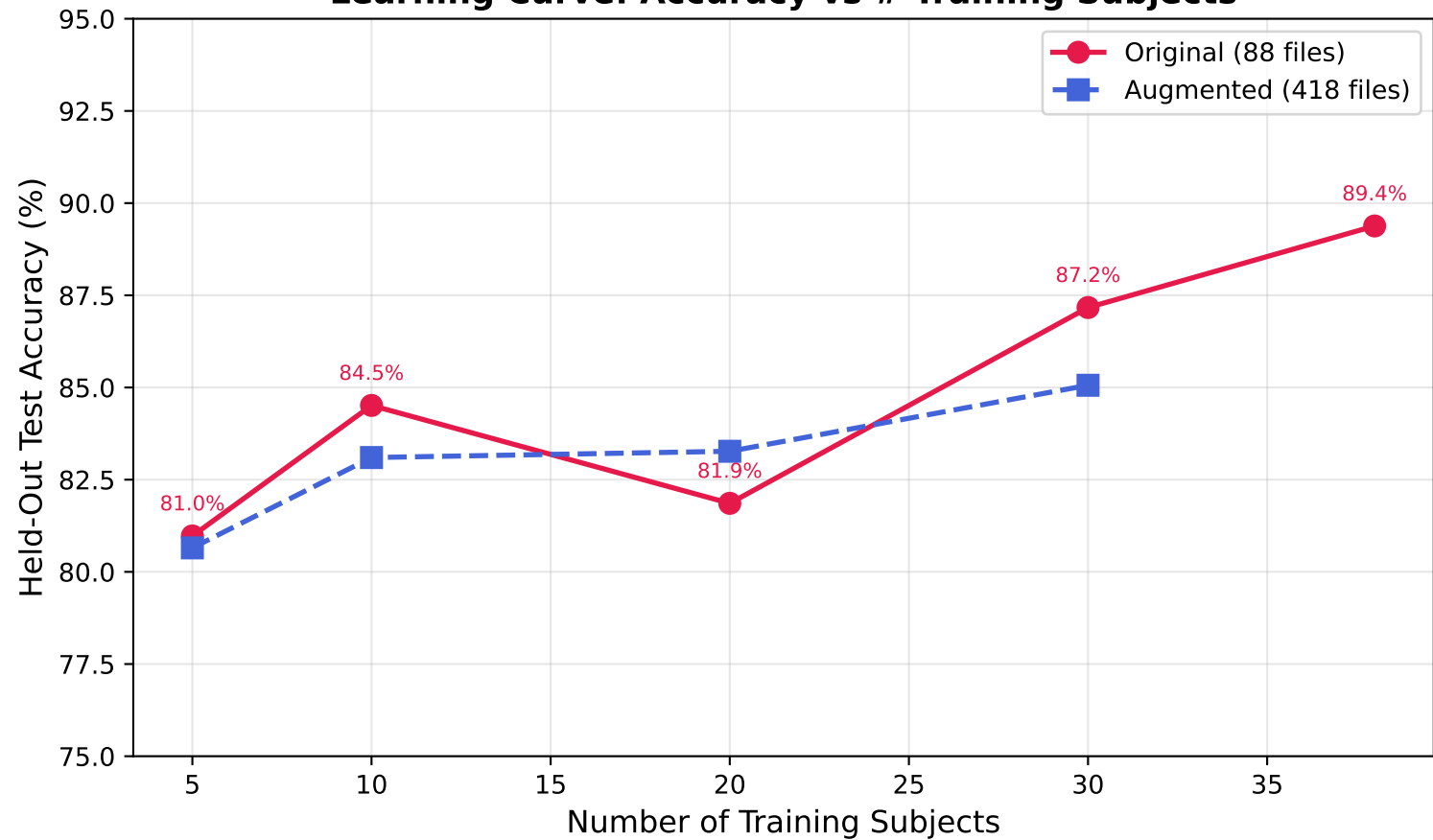
Original (7,068 windows)



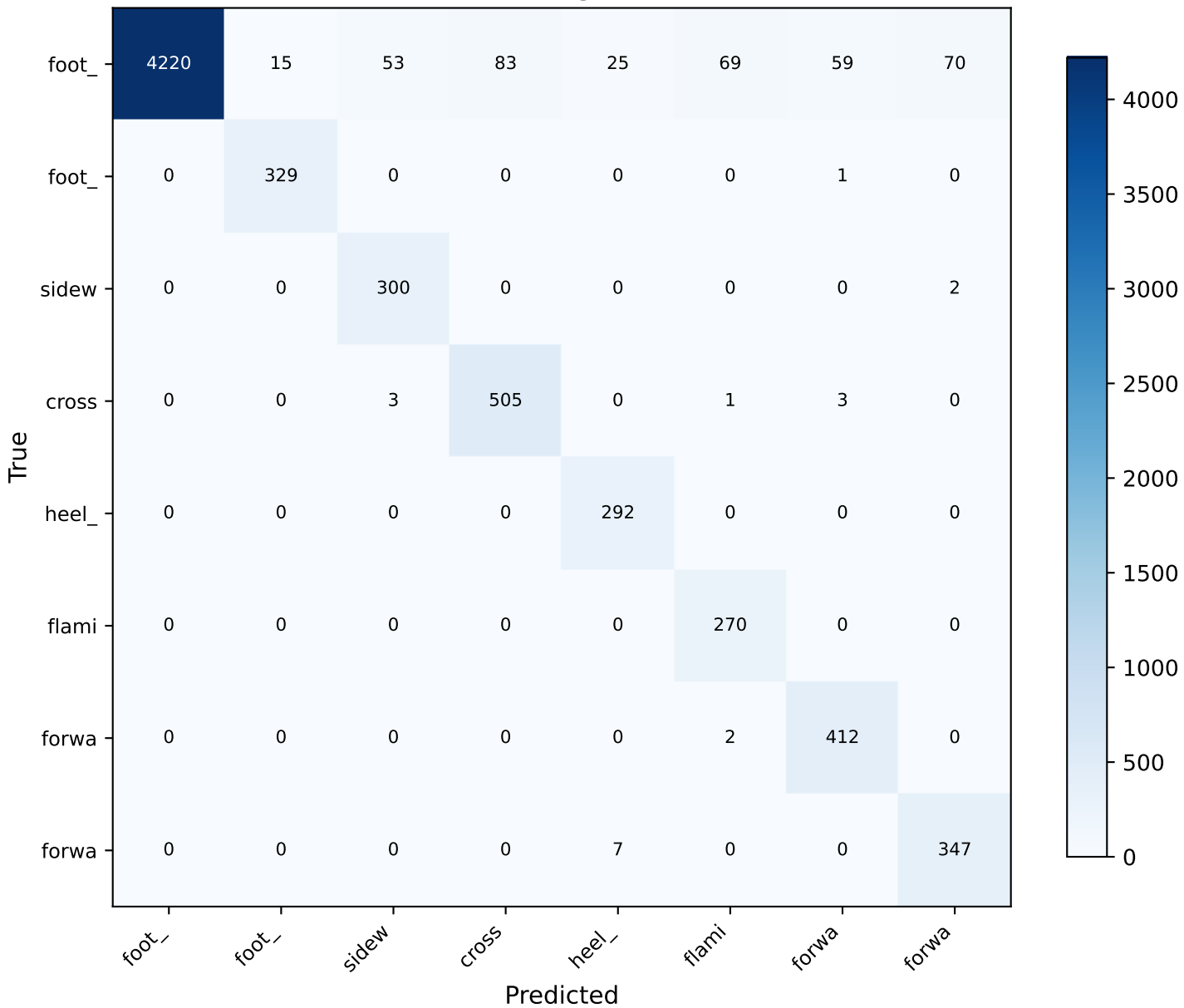
Augmented (44,995 windows)



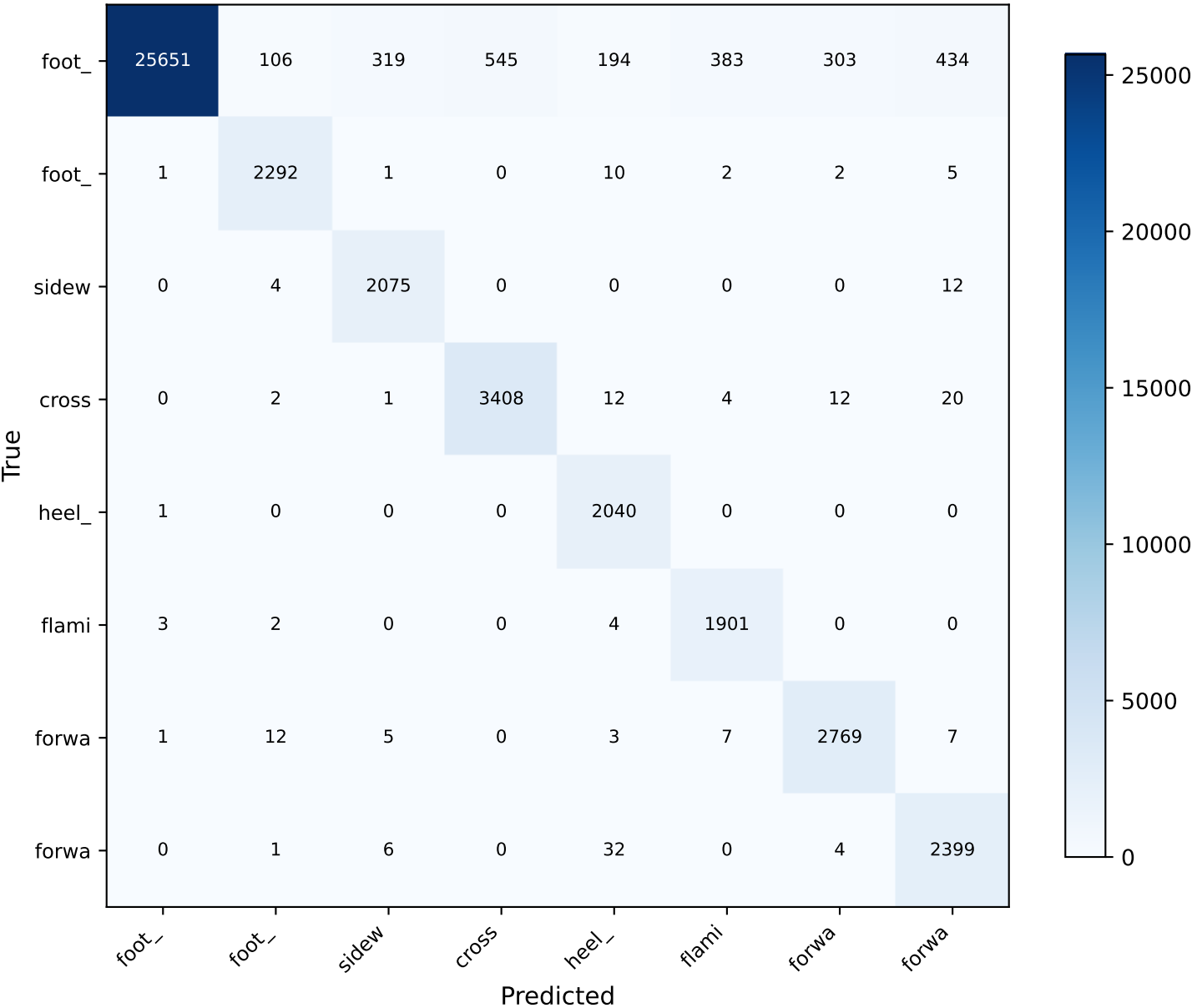
Learning Curve: Accuracy vs # Training Subjects



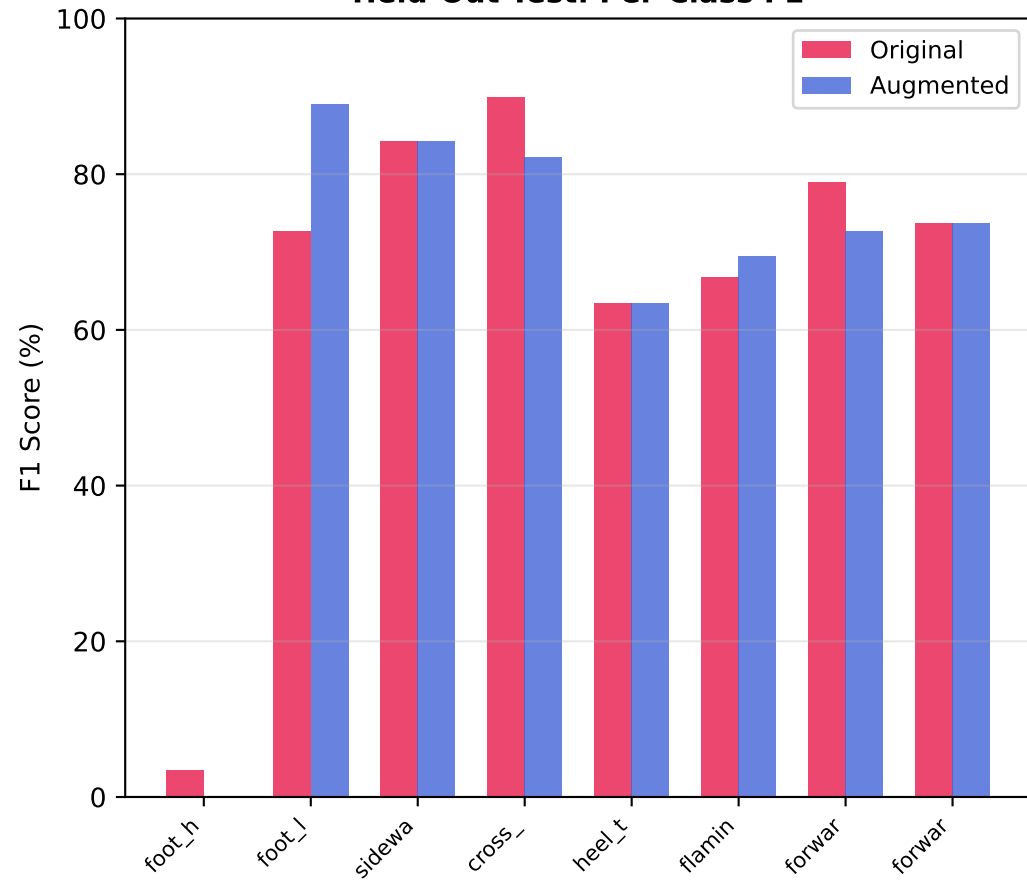
Confusion Matrix — Original Model (94.44%)



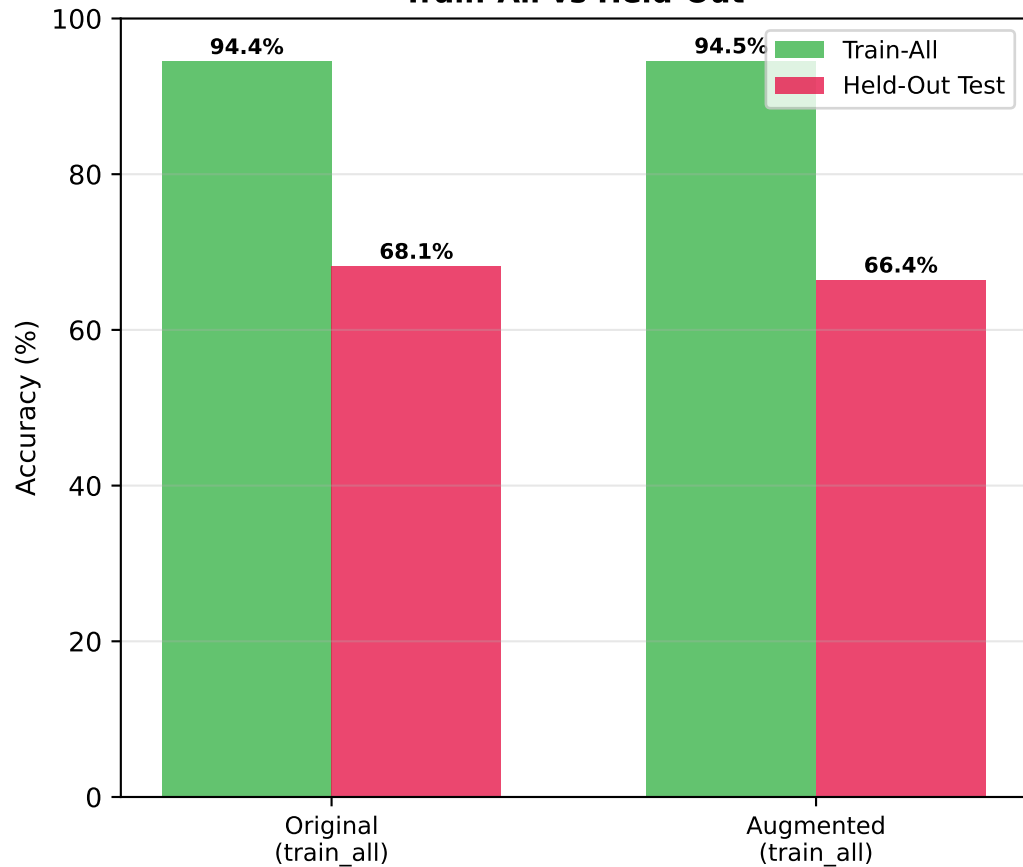
Confusion Matrix — Augmented Model (94.53%)



Held-Out Test: Per-Class F1

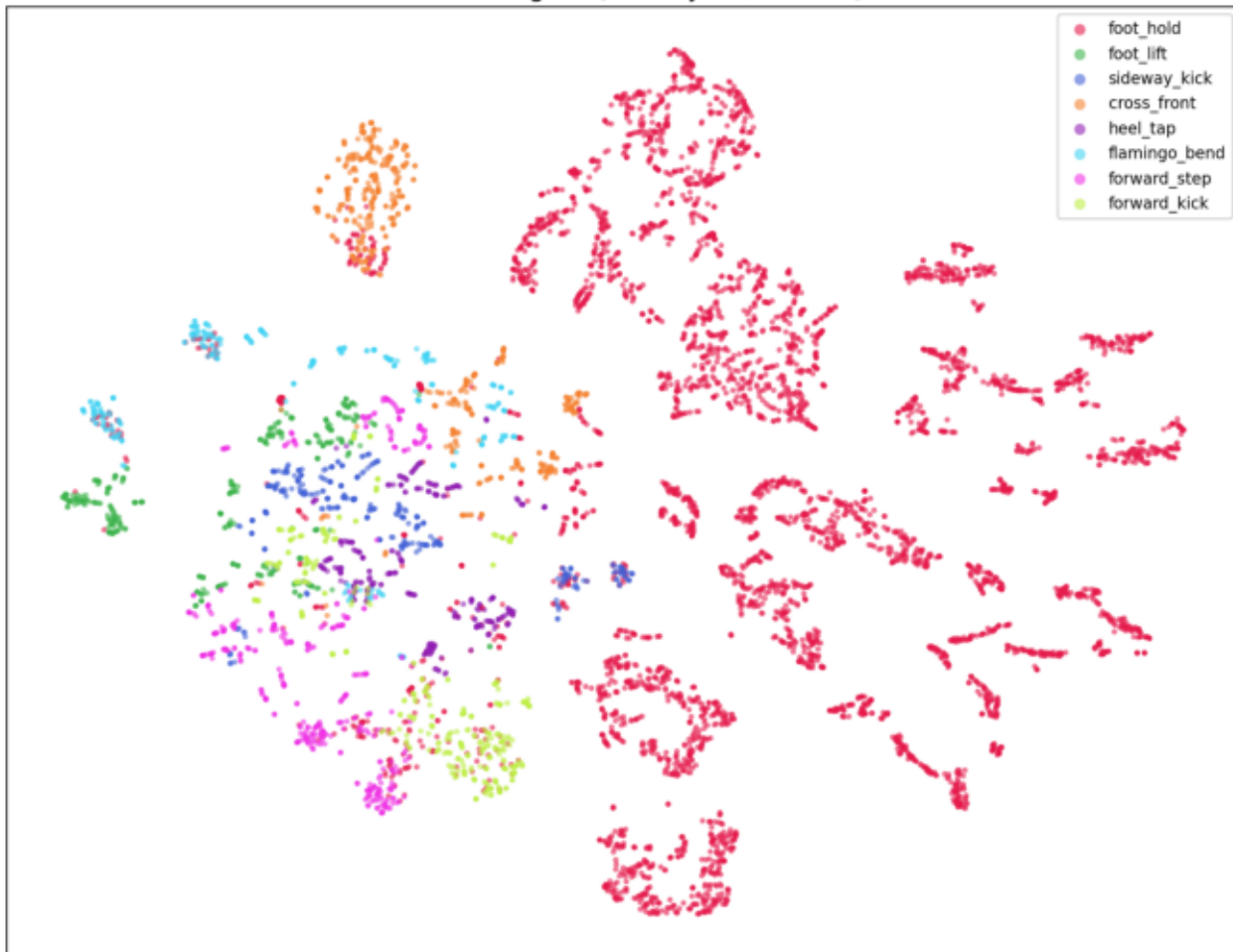


Train-All vs Held-Out



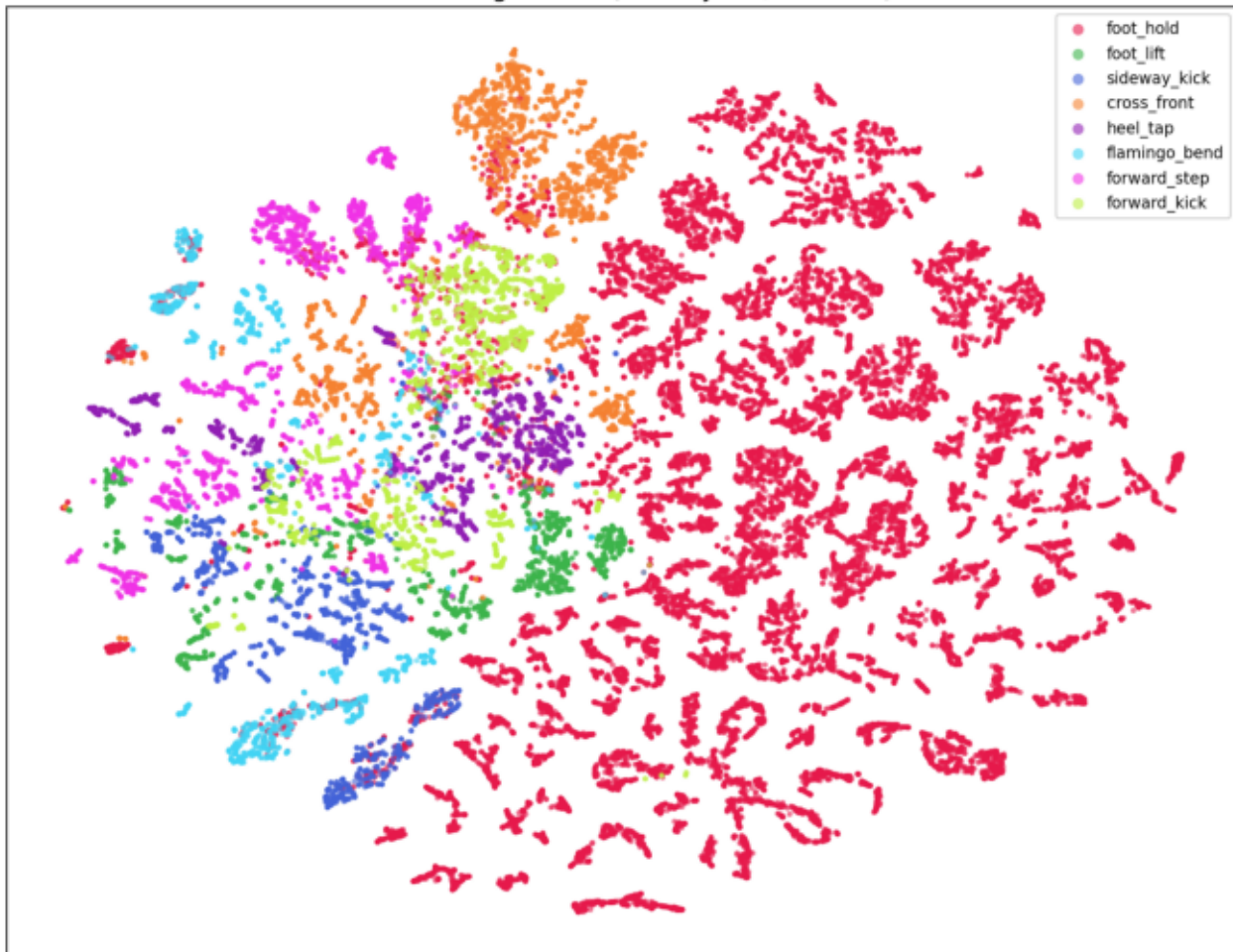
t-SNE Visualization — Original Model

t-SNE — Original (44 subjects, 88 files)



t-SNE Visualization — Augmented Model

t-SNE — Augmented (38 subjects, 418 files)



Ablation Study — All Experiments

#	Config	Dataset	Params	CV Accuracy	Δ
1	Baseline (cnn=64, do=0.6)	front	132K	41.11% $\pm$ 3.59%	—
2	+ cnn_out=128	front	526K	42.80% $\pm$ 3.94%	+1.69
3	+ dropout=0.3	front	132K	43.73% $\pm$ 5.02%	+2.62
4	+ cnn=128 + do=0.3	front	526K	44.36% $\pm$ 6.75%	+3.25
5	BiLSTM baseline	front	207K	24.80%	-16.31
6	BiLSTM + combined	combined	207K	35.12%	-5.99
7	CNN + combined (no flag)	combined	132K	38.29%	-2.82
8	CNN + camera-ID flag	front+45	140K	55.71%	+14.60
9	Final (cnn=128, do=0.3)	front	526K	94.44%*	—
10	+ 5 $\times$ augmentation CV	front_aug	526K	68.96% $\pm$ 22%	-5.27
11	+ 5 $\times$ augmentation train	front_aug	526K	94.53%*	—

KEY FINDINGS

1. CNN-only dominates BiLSTM
+16 pp over BiLSTM baseline (74.10% vs 24.80%)
2. Width matters more than depth
cnn_out=128 + dropout=0.3 → +3.25 pp CV over baseline
3. Multi-camera hurts without flag
38.29% vs 41.11% (front-only). Camera-ID flag: +14.60 pp
4. Augmentation does NOT improve cross-subject generalization
CV: 68.96% vs 74.23% (-5.27 pp)
Held-out: 66.37% vs 68.14% (-1.77 pp)
Synthetic variants don't teach the model new people
5. Data scarcity is the primary bottleneck
Train-all: ~94.5% | Held-out: ~68% | CV mean: ~69-74%
26 pp gap = memorization without generalization
6. foot_hold collapses on unseen subjects
25% precision (original), 0% (augmented)
Clearest overfitting signal
7. Learning curve still climbing at 38 subjects
80.97% (5 subj) → 89.38% (38 subj)
More real subjects = best path forward

RECOMMENDATIONS FOR PAPER

- Report held-out test (68.14%) as headline result
- Show learning curve as evidence of data bottleneck
- Include augmentation as negative result
- Report per-class F1 – foot_hold confusion is key story
- Consider LOSO CV for gold-standard evaluation